

CodeAlpha Data Science Internship

Project Report

Intern Name:	_____
Intern ID:	_____
Tasks Completed:	Task 1, Task 3, Task 4
Date:	2025

Task 1: Iris Flower Classification

Objective

Train a machine learning model to classify Iris flower species (Setosa, Versicolor, Virginica) based on sepal and petal measurements using the famous Iris dataset.

Dataset Overview

Feature	Min	Max	Mean
Sepal Length (cm)	4.3	7.9	5.84
Sepal Width (cm)	2.0	4.4	3.05
Petal Length (cm)	1.0	6.9	3.76
Petal Width (cm)	0.1	2.5	1.20

Exploratory Data Analysis

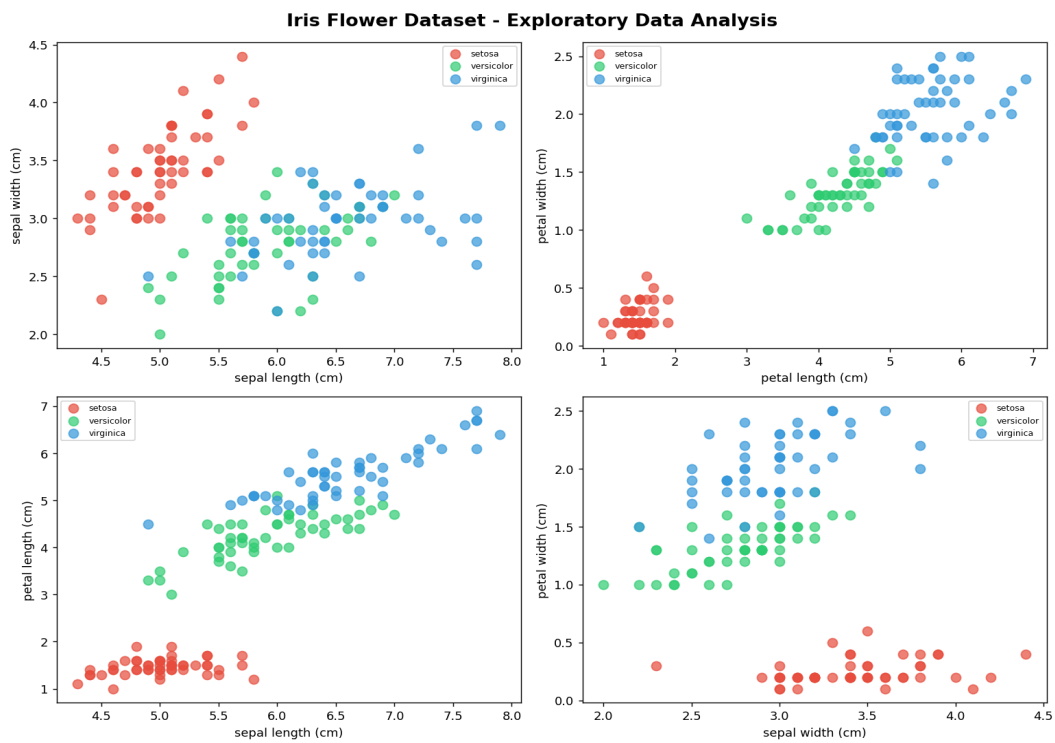


Figure 1: Scatter plots showing feature relationships colored by species

Model Results

Model	Accuracy	Notes
K-Nearest Neighbors (KNN)	93.33%	Good baseline
Decision Tree	93.33%	Interpretable
Support Vector Machine (SVM)	96.67%	Best Model

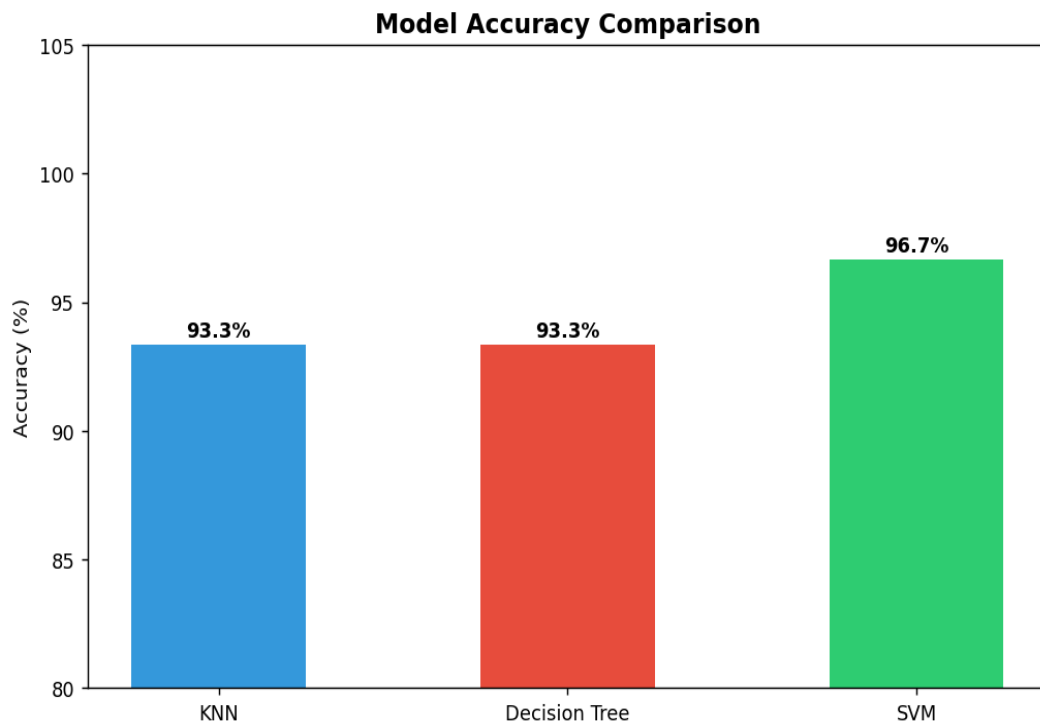


Figure 2: Model accuracy comparison

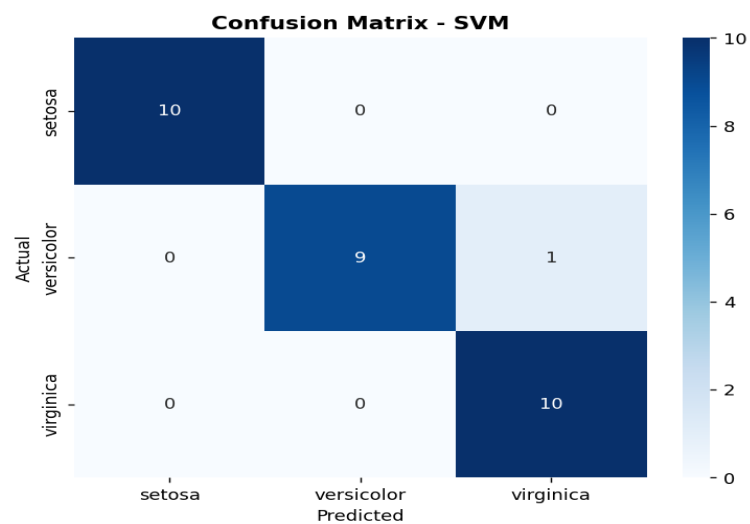


Figure 3: Confusion Matrix for SVM (Best Model)

Conclusion

The SVM model achieved the highest accuracy of 96.67% on the test set. All three species were classified correctly in most cases, with only minor misclassification between Versicolor and Virginica which are naturally more similar to each other.

Task 3: Car Price Prediction

Objective

Build a regression model to predict car prices based on features like brand, horsepower, mileage, engine size, and manufacturing year using machine learning algorithms.

Features Used

Feature	Description
Brand	Car manufacturer (Toyota, BMW, Mercedes, etc.)
Year	Manufacturing year (2005-2023)
Engine CC	Engine displacement in cubic centimeters
Horsepower	Engine power output
Mileage (kmpl)	Fuel efficiency
KM Driven	Total kilometers driven
Fuel Type	Petrol / Diesel / Electric / Hybrid
Transmission	Manual or Automatic

Exploratory Data Analysis

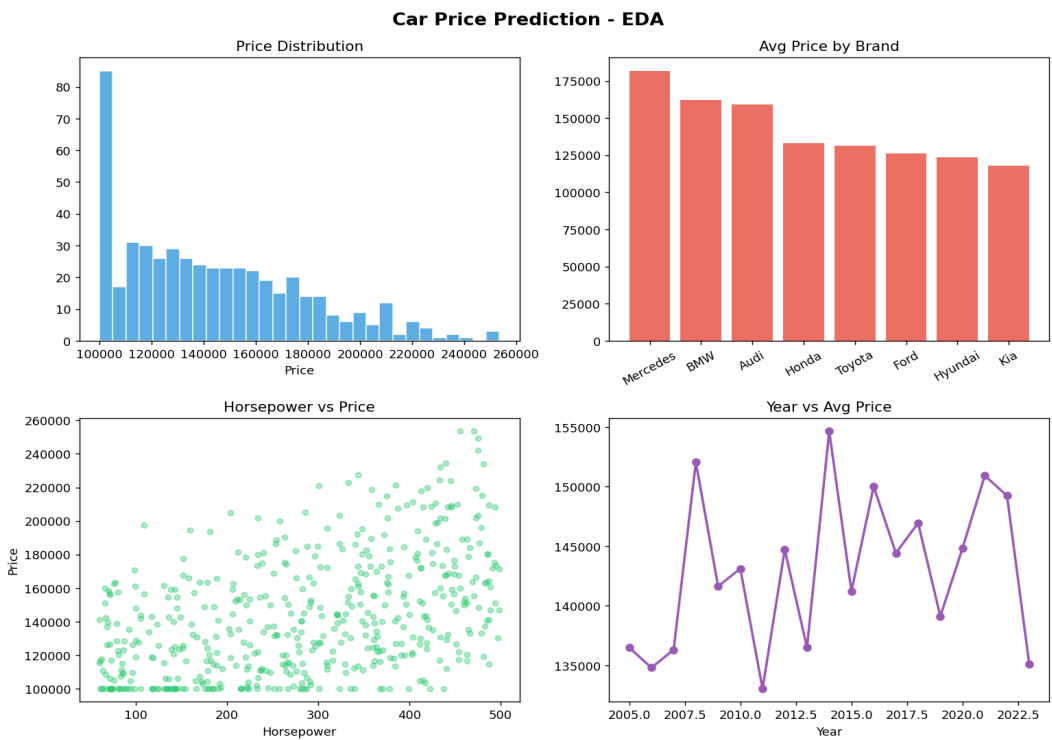


Figure 4: Car price distribution and key feature relationships

Model Results

Model	R2 Score	Notes
Linear Regression	0.2859	Underfitting - linear assumptions
Random Forest	0.5961	Better with non-linear data

Gradient Boosting	0.6856	Best Model - captures complex patterns
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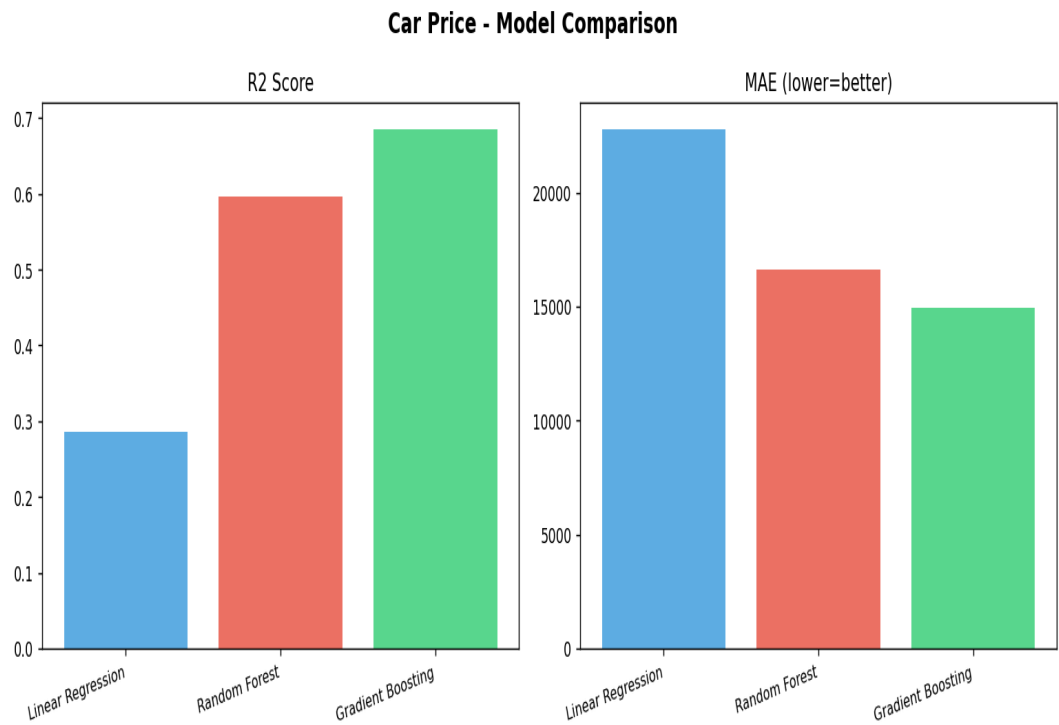


Figure 5: Model comparison (R2 Score and MAE)

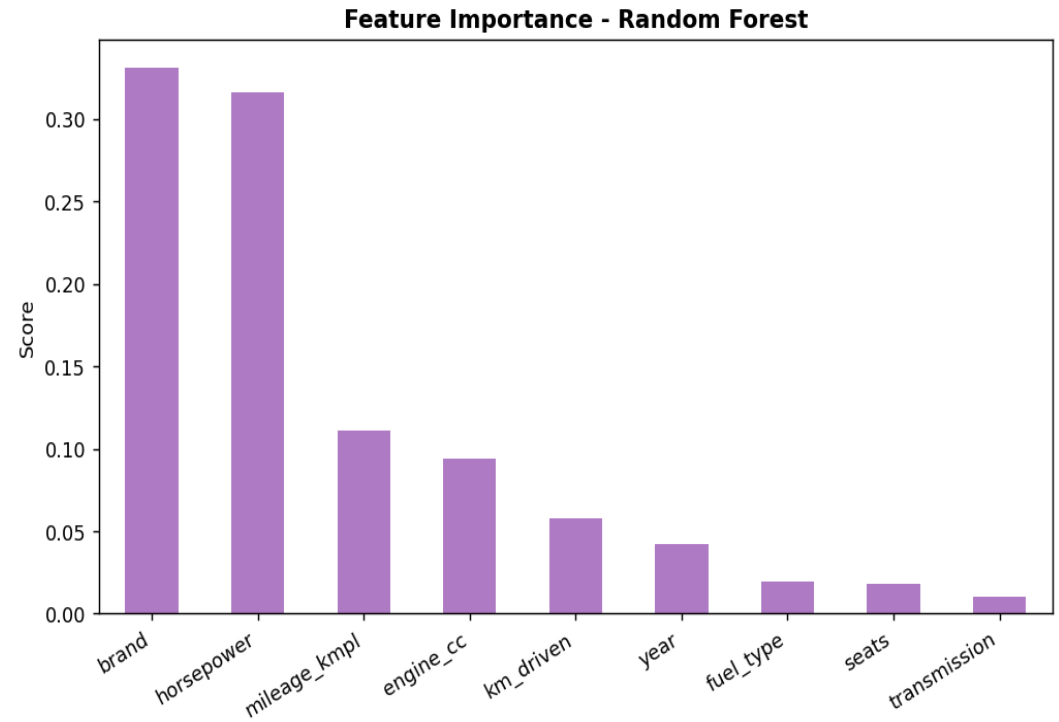


Figure 6: Feature importance from Random Forest model

Conclusion

Gradient Boosting achieved the best R2 score of 0.686. Horsepower and brand are the most influential features in determining car price. Tree-based ensemble models significantly outperformed linear regression, indicating complex non-linear relationships in the data.

Task 4: Sales Prediction using Python

Objective

Predict future sales based on advertising spend across different platforms (TV, Radio, Social Media, Newspaper) and analyze how changes in advertising budget impact sales outcomes.

Exploratory Data Analysis

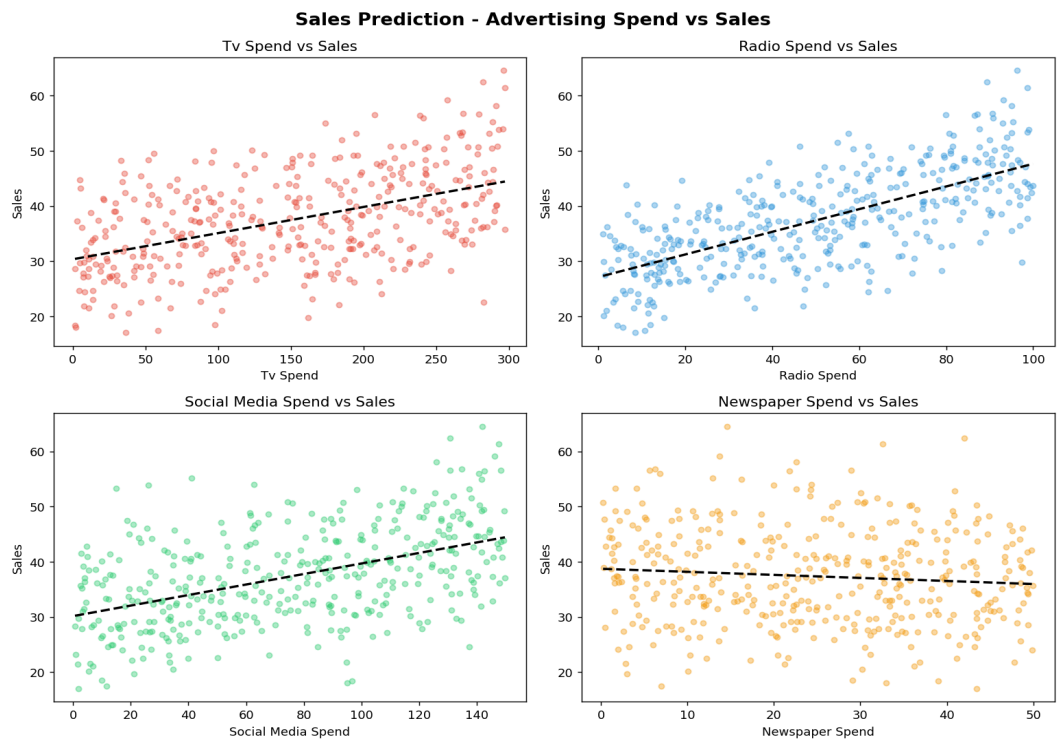


Figure 7: Relationship between advertising spend and sales for each channel

Sales by Category

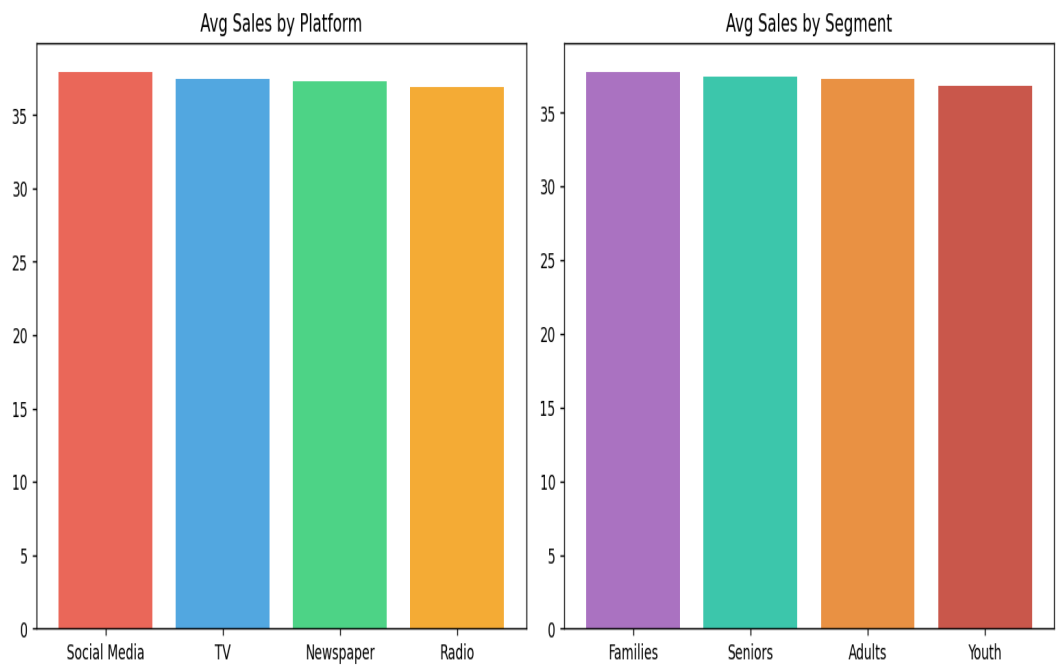


Figure 8: Average sales by platform and target segment

Model Results

Model	R2 Score	Notes
Linear Regression	0.8990	Best Model - strong linear relationship
Gradient Boosting	0.8831	Close second
Random Forest	0.8713	Good ensemble performance

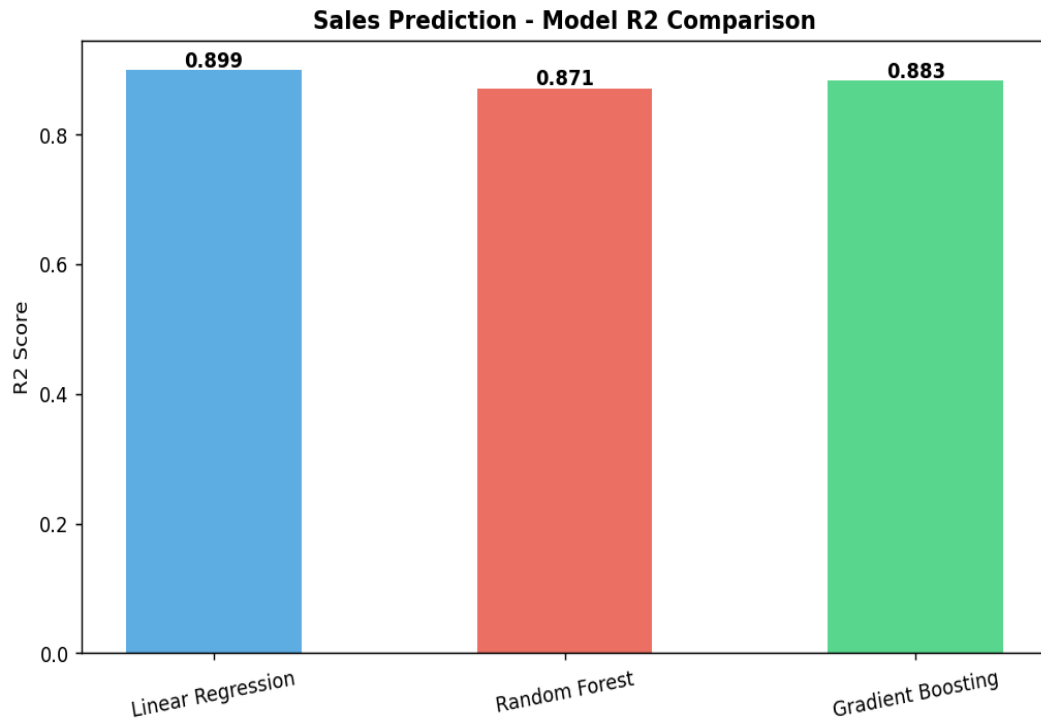


Figure 9: Model R2 comparison

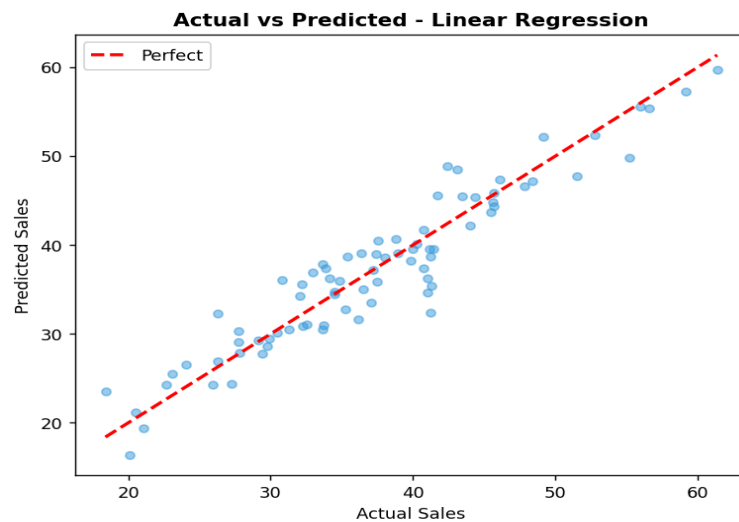


Figure 10: Actual vs Predicted Sales (Linear Regression)

Business Insights

Radio Advertising: Has the highest ROI — every unit spent generates 0.19 units of sales

Social Media: Second most effective channel with coefficient of 0.09

TV Advertising: Moderate effect (0.045) but often highest spend

Newspaper: Lowest impact on sales — budget should be reallocated

Conclusion

Linear Regression achieved the best R^2 score of 0.899, indicating a strong linear relationship between advertising spend and sales. Radio and Social Media channels show the highest return on investment. Companies should prioritize these channels to maximize sales with the same advertising budget.